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Hanna, Alberta, Canada • Remote

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Dear Hiring Committee,

I am writing to express my strong interest in the Contract Faculty position for **CPSC 2050: Human-Computer Interaction** at the University of Fredericton. With a Master of Science in Computer Science from the University of Calgary, more than twenty years of professional software development experience spanning interactive graphics, game engineering, and user-facing web technologies, and active online teaching experience across a full-stack diploma program, I believe I am exceptionally well positioned to deliver a rigorous and engaging HCI course.

Human-computer interaction has been a continuous thread through my entire career, not a single project or credential. My first professional role at **Codefast.com** was to build graphical user interfaces for existing command-line tools—the quintessential HCI task of making complex workflows accessible to a broader audience. At **PlayStarMusic Corp.**, I designed the end-user interface for a media streaming storefront, focusing on browsing flow, playback controls, and seamless DRM integration that preserved the listening experience. At **BJ Pipeline Inspection Services**, I built 3D OpenGL visualizations of pipeline inspection data for professional engineers and independently ported a cumbersome Excel-based reporting workflow into a full web application with interactive JavaScript forms, dynamic charts, and live geographic maps—turning a specialist-only tool into something accessible to non-technical stakeholders across the organization.

At **Fekete Associates**, I contributed to the shared MFC UI tools library that served as the consistent interaction foundation for all internal software products, alongside designing and delivering the OpenGL graphics elective described above.

At **bitHeads Inc.**, I was responsible for building demonstration applications and widget libraries for every supported client platform to showcase SDK integration—a cross-platform UI portfolio spanning Dear ImGui (C++), Unreal Engine UMG, SwiftUI/UIKit, Android Views, Unity, and the PlayStation SDK. Effective developer experience—making an API intuitive, discoverable, and self-documenting—is itself an HCI discipline, and I brought that lens to every interface I built.

My game development work at **Silicon Hanna Inc.** added the consumer product dimension: touch interface design, augmented reality (ARKit), and physics-based feedback loops, all evaluated through the lens of player experience—where the quality of the human-computer interaction is the product itself.

On the teaching side, I have been a Technology Instructor at **Robertson College** since January 2025, delivering online courses including *Front End Frameworks* (Angular and React), *Web Development Basics*, *ASP.NET Core*, and *JavaScript*—courses whose central challenge is teaching students to build interfaces that real users will actually want to use. I conduct a minimum of three synchronous live sessions per week, host live-coding and debugging workshops, and maintain open-source course materials on GitHub. I have earned an average student satis-

faction rating of **4.5 out of 5.0**, with consistent feedback highlighting my approachability, real-world context, and responsiveness.

I am also familiar with the Brightspace (D2L) LMS environment, which I use daily at Robertson College—the same platform the University of Fredericton employs—and I am comfortable with all aspects of asynchronous and synchronous online course delivery.

My academic background includes graduate research in agent-based computing and biologically inspired systems, contributions to five published research papers, and recognition with a departmental nomination for the Governor General's Gold Medal, an NSERC Postgraduate Scholarship (D3), and a Best Student Paper Award at the Australian Conference on Artificial Life. I bring both the technical depth and the scholarly perspective that a university-level HCI course demands.

I would welcome the opportunity to discuss how my background in interactive systems, interface design, and technology education aligns with the goals of CPSC 2050. Thank you for your time and consideration.

Sincerely,

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